

PART 1 — ENGLISH

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-17

Subject: Robotic cathode stripping machine, 120 cathodes/h.

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting-leaching-electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal (budgetary quotation) for the item above. Project site: an operating metallurgical site in north-west Russia.

Brief description of the item: Two-stage washing, stripping, optical inspection with machine vision, electronic register of mother blanks, bundling, automatic sampling; installed power 125 kW. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price quoted separately on EXW (manufacturer's works) and DDP project site in north-west Russia terms (Incoterms 2020), stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (weeks from order / advance payment);
3. overall dimensions and weights of the equipment and of the largest shipping blocks (for logistics studies);
4. reference list of similar supplies: product, process duty and parameters, year of supply, country;
5. materials of parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements.

Annex 1. Technical requirements

To RFQ No. OVE75-RFQ-17 — values follow the Customer's input data and terms of reference; Basic Engineering (FEED) stage, subject to refinement at the next project stage.

RFQ item	Robotic cathode stripping machine, 120 cathodes/h
Summary	Two-stage washing, stripping, optical inspection with machine vision, electronic register of mother blanks, bundling, automatic sampling; installed power 125 kW
Specific requirements	Mandatory requirements: separation of the two deposit sheets without bonding along the bottom edge; bundle "perforation" by profiling of the sheets; lot formation with accompanying documentation

Equipment covered by this request:

1. Robotic cathode stripping machine (CSM) Cathode handling

Type / model	Reference — Outotec supply: KUKA KR 300 (feeding) and KR 180 (discharge) robots
Duty	Full cathode processing cycle: washing, separation of the deposit from the mother blanks, optical inspection and rejection, bundling, automatic sampling
Key parameters	Design throughput 105 cathodes/h, adopted 120 cathodes/h (Outotec reference — 250 mother blanks/h); positioning accuracy up to 1 mm; operating personnel: 1
Quantity	1
Materials	Structural and stainless steel, PVC, polypropylene
Dimensions, hold-up	Installed power 125 kW, consumed power 105 kW; 380 V 50 Hz, controls 220 V; hydraulics 120 bar, pneumatics 6 bar; cooling water 1.5 bar, ≤ 30 °C, 25 L/min; washing stage I: 25 m ³ /h at 60–80 °C, stage II: 4 m ³ /h at 90–115 °C
Additional requirements	Mandatory: separation of the two deposit sheets without bonding along the bottom edge; electronic register of the mother blank fleet; bundle "perforation" by profiling of the sheets; bundles of 1,400–2,000 kg (sales requirement — $2,500 \pm 50$ kg, strapped with three corrosion-resistant bands)

2. Cathode bundle forming equipment Cathode handling

Duty	Stacking, weight control, marking, strapping, bundle "perforation"
-------------	--

Key parameters	Bundles of 1,400–2,000 kg (sales requirement: 2,500 ± 50 kg)
Quantity	Complete set

3. Automated transfer of copper cathode bundles to the finished goods warehouse **Materials handling**

Duty	Transfer of the bundles to the finished goods warehouse
Quantity	Complete set

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

PART 2 — 中文 (CHINESE)

预算性技术商务报价询价函 编号 OVE75-RFQ-17

事由：机器人阴极剥片机组，120 片/小时。

KVANT LLC（俄罗斯联邦）正在开展铜冶炼厂建设项目的基础设计（FEED）——“焙烧—浸出—电积”联合工艺，年产阴极铜 75,000 吨——现请贵司就上述设备提交预算性技术商务报价（预算报价）。项目场地：俄罗斯西北部一座在运营的冶金厂区。

设备简要说明：两段洗涤、剥片、机器视觉光学检测、母板电子台账、打捆、自动取样；装机功率 125 kW。技术要求见本询价函附件 1。

请在报价中包含：

1. 预算价格——分别按 EXW（制造商工厂）及 DDP 至俄罗斯西北部项目场地（Incoterms 2020）条件报价，注明币种及报价有效期；
2. 制造周期及预计交货期（自订单/预付款之日起按周计）；
3. 设备单机及最大运输单元的外形尺寸和重量（用于物流研究）；
4. 类似供货业绩表：产品、工况介质及参数、供货年份、国家；
5. 与工艺介质接触部件的材质，以及供货范围（仪表、备件、技术文件）；
6. 公用工程消耗及安装要求（如有资料）。

本询价为基础设计（FEED）阶段的预算性询价：不构成要约，双方均不因此承担义务；详细技术数据表将在项目下一阶段发出。

回复期限：请在收到本询价函后 3（三）周内将报价发送至 stepan@kvantpro.com。请确认收到本询价函。

保密条款。本询价函及随附技术资料仅供编制报价使用，属保密信息，未经 KVANT LLC 事先书面同意，不得向第三方披露、复制或发布。提交报价不为双方设立任何法律义务。

联系人：Stepan Kharkov, KVANT LLC, stepan@kvantpro.com。

附件 1：本询价设备的技术要求。

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-17 — 参数依据业主原始数据及设计任务书；基础设计（FEED）阶段，后续阶段可能进一步细化。

询价设备	机器人阴极剥片机组，120 片/小时
简要说明	两段洗涤、剥片、机器视觉光学检测、母板电子台账、打捆、自动取样；装机功率 125 kW
特殊要求	强制性要求：两张沉积铜板沿下边缘不得粘连、须可分离；通过板材压型实现铜捆“孔隙化”；按批次成组并附随行文件

本询价涵盖的设备清单：

1. 机器人阴极剥片机（CSM） 阴极处理

型号 / 参考	参考业绩——Outotec 供货：KUKA KR 300（上料）与 KR 180（下料）机器人
用途	阴极全流程处理：洗涤、阴极铜与母板分离（剥片）、光学检测与剔除、打捆、自动取样
主要参数	设计能力 105 片/小时，采用 120 片/小时（Outotec 参考业绩——250 片母板/小时）；定位精度达 1 mm；操作人员 1 名
数量	1
材质	结构钢与不锈钢、PVC、聚丙烯
尺寸、滞留量	装机功率 125 kW，耗用功率 105 kW；380 V 50 Hz，控制电源 220 V；液压 120 bar，气动 6 bar；冷却水 1.5 bar，≤30 °C，25 L/min；洗涤水一段 25 m³/h、60–80 °C，二段 4 m³/h、90–115 °C
附加要求	强制性要求：两张沉积铜板沿下边缘不得粘连、须可分离；母板电子台账；通过板材压型实现铜捆“孔隙化”；铜捆重 1,400–2,000 kg（按销售要求—— $2,500 \pm 50$ kg，用三道耐腐蚀捆带捆扎）

2. 阴极铜捆成型设备 阴极处理

用途	堆垛、称重、标识、捆扎、铜捆“孔隙化”
主要参数	铜捆 1,400–2,000 kg（销售要求 $2,500 \pm 50$ kg）
数量	成套

3. 阴极铜捆至成品库的自动化输送 起重运输设备

用途	将铜捆输送至成品库
数量	成套

--	--

报价中须确认：工况介质、温度、压力、接触部件材质、生产能力、外形尺寸及重量、能耗、制造周期、交货条件。